

4주차 미적과제

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$$1. f'(2) = 4 \quad \frac{f(4) - f(1)}{4 - 1} = f'(c) \rightarrow \frac{f(4) - 10}{3} = 4 \quad \boxed{f(4) = 22}$$

$$2. f'(x) = 2x - 1 - \frac{1}{x} \quad x=1 \quad (x > 0)$$

정의역: $[1, \infty)$ 극값: X
 구간: $[0, 1]$ 극값: $f(1) = 0$

$$f''(x) = 2 + \frac{1}{x^2} \neq 0 \quad \text{변곡점: X}$$

$$3. \lim_{x \rightarrow 0} 0 \rightarrow x=2, x=-2 \quad \text{극점: } x=2, x=-2 \quad f(x) = \frac{x}{x^2-4}$$

$$\lim_{x \rightarrow \infty} f(x) = 0 \quad \lim_{x \rightarrow -\infty} f(x) = 0 \quad \text{수렴값: } y=0$$

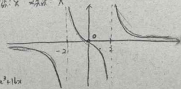
$$f(x) = \frac{x^2-4}{(x^2-4)^2} = \frac{-(x^2-4)}{(x^2-4)^2} < 0 \quad \text{정의역: } x \quad \text{구간: } (-\infty, -2), (-2, 2), (2, \infty)$$

극값: X 변곡점: X

$$-x^2-4 \rightarrow -2x \quad \frac{-(x^2-4)}{(x^2-4)^2} > 0$$

$$(x^2-4)^2 \rightarrow 4x^2-16x \quad \frac{-(x^2-4)}{(x^2-4)^2} < 0$$

$$f''(x) = \frac{-2x(x^2-4)^2 + (x^2-4)(4x^2-16x)}{(x^2-4)^4} = \frac{-2x^3+8x+4x^2-16x}{(x^2-4)^3}$$



$$= \frac{2x^3+24x}{(x^2-4)^3} = \frac{2x(x^2+12)}{(x^2-4)^3} \quad x=0 \text{ 일 때 변곡점 } (0,0)$$

$$4. (1) \lim_{x \rightarrow 1} \frac{x^2+x-2}{x-1} = \frac{(x-1)(x+2)}{(x-1)} = (x+2) = 3$$

$$(2) \lim_{x \rightarrow 0} \frac{\cos x - 1}{x^2} \rightarrow \frac{-2\ln 2x + \sin x}{2x} \rightarrow \frac{-4\cos 2x + \cos x}{2} = -\frac{3}{2}$$

$$(3) \lim_{x \rightarrow 0^+} \left(\cot x - \frac{1}{x} \right) \rightarrow \left(\frac{-1}{1+x^2} + \frac{1}{x} \right) = \frac{1}{x^2(x^2+1)} \rightarrow \infty$$

$$(4) \lim_{n \rightarrow \infty} [2 - 2^n \ln(\frac{1+x}{x})] \quad x = \frac{1}{n} \quad \lim_{n \rightarrow 0} \left[\frac{1}{n} - \frac{\ln(1+\frac{1}{n})}{\frac{1}{n^2}} \right] \rightarrow \frac{1 - \frac{1}{n+1}}{2n} = \frac{n}{2n(n+1)} = \frac{1}{2}$$